MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

Guideline for the Treatment of Cryptococcosis in Adults

Infection Type & Pathogens	Management of Cryptococcal infection (ID consult is required)		Comments
Pulmonary Cryptococcosis ***Cryptococcus neoformans*** ***Cryptococcus gattii***	**Mild**	With or without pulmonary cryptococcoma: **Fluconazole** 400-800 mg* PO daily x 6-12 months	• Isolation of *Cryptococcus* spp in cultures from any site should not be considered colonization and ID consultation should be obtained • Mild pulmonary disease is defined as asymptomatic or mildly symptomatic patients or those with a solitary small nodule (< 2 cm) • Severe disease is defined as multiple lesions, large lesions (≥ 2 cm), lobar consolidation, cavitation, multi-lobar involvement, or hypoxemia • Exclusion of an underlying immunodeficiency (including performing HIV serology and CD4 T-cell count) is recommended in all patients with cryptococcosis • The use of liposomal amphotericin B should be accompanied by pre-hydration and aggressive potassium and magnesium replacement therapy
	Severe: Induction	Without pulmonary cryptococcoma: **Liposomal Amphotericin B** 3 mg/kg IV daily ×14 days **+** **Flucytosine** 25 mg/kg* PO QID ×14 days With pulmonary cryptococcoma: **Liposomal Amphotericin B** 3 mg/kg IV daily × 4-6 weeks **+** **Flucytosine** 25 mg/kg* PO QID × 4-6 weeks Salvage regimens based on tolerability which may be suboptimal; Consult ID pharmacy prior to adjustment: *Liposomal amphotericin B intolerance:* **Fluconazole** 1200 mg/day* with **flucytosine** 25 mg/kg* PO QID *Flucytosine intolerance:* **Liposomal amphotericin B** 3mg/kg IV daily with **fluconazole** 800 mg/day* *Liposomal amphotericin B AND flucytosine intolerance*: **Fluconazole** 1200 mg/day*	
	Severe: Consolidation	**Fluconazole** 400-800 mg* PO daily x 8 weeks	
	Severe: Maintenance	**Fluconazole** 200 mg* PO daily x1 year or until immune restoration (in patients with HIV, defined as undetectable viral load on ART with CD4> 100 cells/mm^3)	

<table>
<tr><td rowspan="5">Cryptococcal Meningitis

Cryptococcus neoformans

Cryptococcus gattii</td><td>Induction</td><td>Preferred in patients without C. gattii or cryptococcoma:
Liposomal Amphotericin B 3 mg/kg IV daily ×14 days
+
Flucytosine 25 mg/kg* PO QID ×14 days

Preferred in patients with C. gatti or cryptococcoma:
Liposomal Amphotericin B 3 mg/kg IV daily × 4-6 weeks
+
Flucytosine 25 mg/kg* PO QID × 4-6 weeks

Salvage regimens based on tolerability which may be suboptimal; Consult ID pharmacy prior to adjustment:
Liposomal amphotericin B intolerance:
Fluconazole 1200 mg/day* with flucytosine 25 mg/kg* PO QID

Flucytosine intolerance:
Liposomal amphotericin B 3 mg/kg IV daily with fluconazole 800 mg/day*

Liposomal amphotericin B AND flucytosine intolerance:
Fluconazole 1200 mg/day*

Alternative regimen only under ID attending guidance in patients living with HIV:
Liposomal Amphotericin B 10 mg/kg IV ×1 dose
+
Flucytosine 25 mg/kg* PO QID ×14 days
+
Fluconazole 1200 mg* PO daily x14 days</td><td rowspan="3">• Morbidity and mortality in Cryptococcal meningitis are mainly driven by complications related to intracranial pressure.
• Patients on liposomal amphotericin should remain inpatient for management of complications from induction therapy.
• The use of liposomal amphotericin B should be accompanied by pre-hydration and aggressive potassium and magnesium replacement therapy.
• Repeat LP is recommended at the end of 2 weeks of induction therapy to assess CSF sterility and opening pressure; LP may need to be repeated sooner if there is concern for ICP elevation.
• Induction therapy should be extended if CSF cultures at 2 weeks remain positive or if patient has indications for longer induction therapy
• New initiation of ART should be deferred for 4-6 weeks in patients living with HIV; patients already on ART should be continued on their ART regimen.
• Dexamethasone was shown to be associated with more adverse outcomes in HIV-associated cryptococcal meningitis. Risk benefits should be carefully considered before use. [Beardsley J et al. NEJM 2016]
• Acetazolamide has been shown to be associated with more serious events without clinical benefit for ICP management. Risk benefits should be carefully considered before use. [Newton PN et al. CID 2002]
• Fluconazole is a potent CYP2C9 and CYP2C19 inhibitor and a moderate inhibitor of CYP3A4 and drug-drug interactions should be checked prior to use.
• The use of follow-up blood or CSF cryptococcal antigen (including monitoring of titers) for clinical decision making is discouraged.
• In patients with HIV, restart maintenance therapy if CD4 count drops to less than 100 cells/mm^3</td></tr>
<tr><td>Consolidation</td><td>Fluconazole 800 mg* PO daily x8 weeks</td></tr>
<tr><td>Maintenance</td><td>Fluconazole 200 mg* PO daily x1 year and immune restoration (in patients with HIV, defined as undetectable viral load on ART with CD4 >100 cells/mm^3)</td></tr>
<tr><td>Monitoring and Managing Intracranial Pressure (ICP)</td><td colspan="2">• Suspected cryptococcal meningitis cases should have an initial opening pressure measured.
• For patients with opening pressure greater than ≥ 25 cm H_2O or symptoms of raised ICP, serial lumbar puncture is preferred to reduce the opening pressure to < 20 cm H_2O or 50% of the initial opening pressure until opening pressure is consistently less than 25 cm H_2O.
• Patients with clinical signs of elevated ICP without evidence of cerebral herniation should undergo urgent lumbar puncture with consideration of urgent repeat imaging.
• Patients who suffer rapid neurological deterioration with impending or suspected cerebral herniation (see Appendix I) may require immediate measures other than a lumbar puncture to prevent or treat cerebral herniation, including an external ventricular drain. Such patients should be urgently moved to an ICU setting for further management.</td></tr>
<tr><td></td><td>Useful Internal resources</td><td colspan="2">• *Antifungal renal dosing guidelines
• *Antifungal therapeutic drug monitoring guidelines
• Inpatient interventional procedure roadmap to initiate off-hours lumbar puncture</td></tr>
</table>

Appendix I: Signs and symptoms of elevated intracranial pressure (ICP)

Symptoms of elevated intracranial pressure (ICP) include:

- Headache: often severe, persistent, and worse in the morning or with a change in position.
- Nausea and vomiting: especially in the morning or when the headache is severe.
- Vision changes: blurred or double vision, transient visual obscurations, or a "graying out" of vision.
- Papilledema: swelling of the optic disc due to increased pressure, detected during an eye exam.
- Altered level of consciousness: drowsiness, confusion, or lethargy.
- Balance and coordination issues: difficulty walking, clumsiness, or ataxia.
- Personality or cognitive changes: irritability, memory loss, or difficulty concentrating.

Note that symptoms may vary depending on the underlying cause and severity of the elevated ICP. Clinical signs of impending herniation include:

- Rapid deterioration in level of alertness,
- Significant pupillary asymmetry, unilateral or bilateral fixed and dilated pupils,
- Decorticate (flexion) or decerebrate (extension) posturing,
- Respiratory depression,
- "Cushing triad" of hypertension, bradycardia, and irregular respiration.
- Seizures: Patients who suffer herniation may develop reflexive movements that mimic seizures, and seizures may increase ICP and worsen herniation. This is a neurological emergency.

Appendix II. Recommended first line antifungal therapy by cryptococcosis syndrome (adapted from Chang et al. *Lancet Infect Dis* 2024)

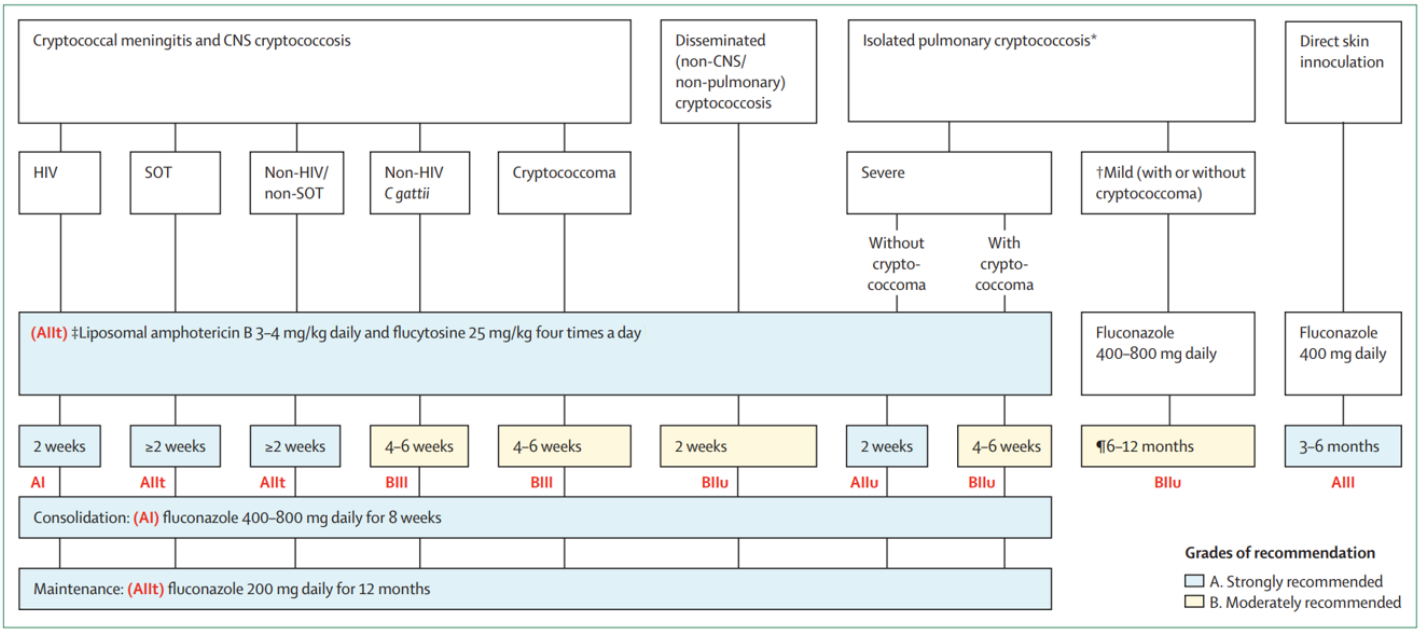


Figure 1: **Recommended first-line antifungal therapy by cryptococcosis syndrome**
Grading of recommendation and level of evidence in bolded red letters. Recommendation grading by shading: blue (strongly recommended; A) and yellow (moderately recommend; B). SOT=solid organ transplantation. *C gattii*=*Cryptococcal gattii*. w=weeks. *Isolated *Cryptococcal neoformans* or *Cryptococcal gattii* pulmonary cryptococcosis, mild is defined as asymptomatic or mildly symptomatic patients or with a solitary small nodule (<2 cm); whereas severe is defined as multiple lesions, large lesions (≥2 cm), lobar consolidation, cavitation, multi-lobar involvement, or hypoxaemic. †If the presence of *Cryptococcus* spp in respiratory specimens is deemed to be airway colonisation after careful evaluation and no treatment is elected, regular follow-up is recommended, especially in the setting of future immunosuppression. ‡Strongly preferred in cryptococcal meningitis and CNS cryptococcosis in SOT and non-HIV non-SOT patient populations, disseminated cryptococcosis, and severe pulmonary cryptococcosis. ‡Has not been directly compared against §. §Has only been trialled in people with cryptococcal meningitis and there are no supporting data of its use in SOT or non-HIV non-SOT patients or in other cryptococcosis syndromes. ¶Can consider a shorter duration (eg, 3 months) in immunocompetent individuals with mild isolated pulmonary cryptococcosis.

References	
	Citation
1	Perfect JR, Dismukes WE, Dromer F, et al. Clinical Practice Guidelines for the Management of Cryptococcal Disease: 2010 Update by the Infectious Diseases Society of America. *Clinical Infectious Diseases*. 2010;50(3):291-322.
2	Guidelines for diagnosing, preventing and managing cryptococcal disease among adults, adolescents and children living with HIV. https://www.who.int/publications-detail-redirect/9789240052178
3	van der Horst CM, Saag MS, Cloud GA, et al. Treatment of cryptococcal meningitis associated with the acquired immunodeficiency syndrome. National Institute of Allergy and Infectious Diseases Mycoses Study Group and AIDS Clinical Trials Group. *N Engl J Med*. 1997;337(1):15-21.
4	Jarvis JN, Lawrence DS, Meya DB, et al. Single-Dose Liposomal Amphotericin B Treatment for Cryptococcal Meningitis. *N Engl J Med*. 2022;386(12):1109-1120.
5	Molloy SF, Kanyama C, Heyderman RS, et al. Antifungal Combinations for Treatment of Cryptococcal Meningitis in Africa. *New England Journal of Medicine*. 2018;378(11):1004-1017.
6	Pappas PG. Managing cryptococcal meningitis is about handling the pressure. *Clin Infect Dis*. 2005;40(3):480-482
7	Beardsley J, Wolbers M, Kibengo FM, et al. Adjunctive Dexamethasone in HIV-Associated Cryptococcal Meningitis. *N Engl J Med*. 2016;374(6):542-554.
8	Newton PN, Thai LH, Tip NQ, et al. A randomized, double-blind, placebo-controlled trial of acetazolamide for the treatment of elevated intracranial pressure in cryptococcal meningitis. *Clin Infect Dis*. 2002;35(6):769-772.
9	Chang CC, Harrison TS, Bicanic TA et al. Global guideline for the diagnosis and management of cryptococcosis: an initiative of the ECMM and ISHAM in cooperation with the ASM. *Lancet Infect Dis.* 2024;24(8):e495-e512.

Revision History	
Date	**Revision**
7/2026	Addition of pulmonary cryptococcosis syndromes and appendix II

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time-to-time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source